

Basic Information

Experiment No.: 07

Date: 08-10-2026

Title: Java ArrayDeque and Its Methods

Objective / Aim

To implement "ArrayDeque" in Java and demonstrate insertion, deletion, and accessing elements from both ends of the deque.

System Requirements

Hardware

- Computer/Laptop
- Minimum 4 GB RAM

Software

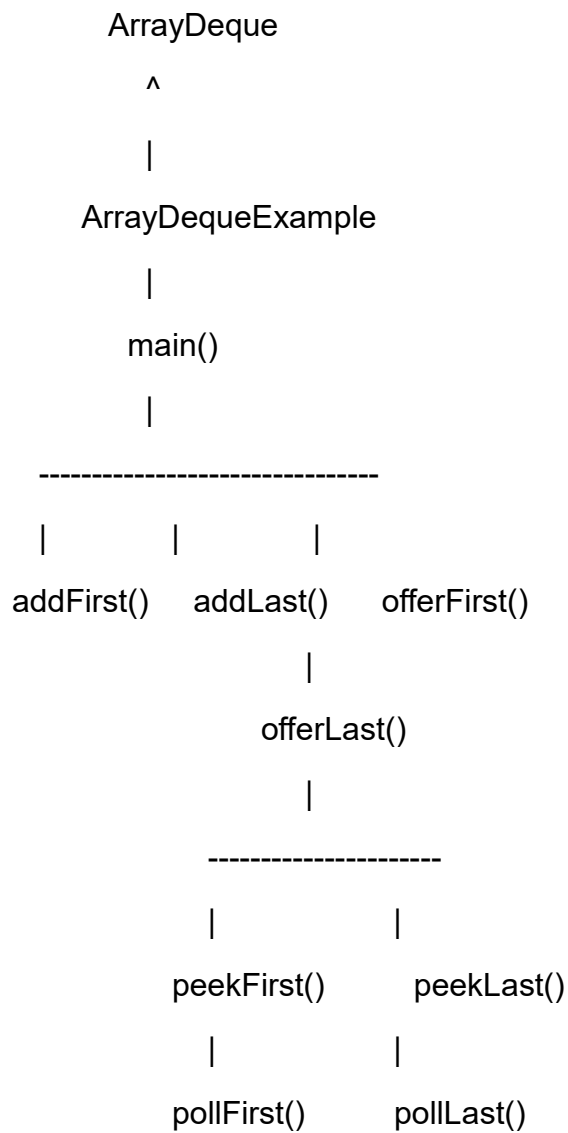
- Java JDK
 - Any Java IDE or Command Prompt
 - Operating System: Windows/Linux/macOS
-

Algorithm / Logic Description

1. Create an "ArrayDeque" of strings.
2. Insert elements using "addFirst()" and "addLast()".
3. Insert elements using "offerFirst()" and "offerLast()".
4. View the first and last elements using "peekFirst()" and "peekLast()".
5. Remove the first and last elements using "pollFirst()" and "pollLast()".

6. Display the deque after each operation.
 7. End the program.
-

UML Class Diagram



Source Code

```
package collectionframework;
```

```
import java.util.ArrayDeque;

public class ArrayDequeExample {
    public static void main(String[] args) {

        ArrayDeque<String> dq = new ArrayDeque<>();

        // addFirst() / addLast()
        dq.addFirst("B");
        dq.addLast("C");
        dq.addFirst("A");
        dq.addLast("D");

        System.out.println("Deque: " + dq);

        // offerFirst() / offerLast()
        dq.offerFirst("Start");
        dq.offerLast("End");

        System.out.println("After offer: " + dq);

        // peekFirst() / peekLast()
        System.out.println("First: " + dq.peekFirst());
        System.out.println("Last: " + dq.peekLast());

        // pollFirst() / pollLast()
        System.out.println("Removed First: " + dq.pollFirst());
        System.out.println("Removed Last: " + dq.pollLast());
```

```
        System.out.println("Final Deque: " + dq);  
    }  
}
```

Compilation & Execution Commands

Save the file as:

ArrayDequeExample.java

Since the program contains:

package collectionframework;

the file should be inside the "collectionframework" folder.

Compile:

javac ArrayDequeExample.java

Run:

java collectionframework.ArrayDequeExample

Output Screen

Deque: [A, B, C, D]

After offer: [Start, A, B, C, D, End]

First: Start

Last: End

Removed First: Start

Removed Last: End

Final Deque: [A, B, C, D]

Result

The "ArrayDeque" class was successfully implemented, and the "addFirst()", "addLast()", "offerFirst()", "offerLast()", "peekFirst()", "peekLast()", "pollFirst()", and "pollLast()" operations were successfully demonstrated.